Hung-Ting Su

hungtingsu.tw@gmail.com • Personal Website • Google Scholar • DBLP • LinkedIn

Robot Learning • Embodied AI • VLA / World Models • Real-World Manipulation

Summary

- Researcher-engineer building reliable embodied AI systems for real-world robot manipulation and navigation, spanning multimodal sensing, VLA-style policies, rollout evaluation, and failure-driven improvement.
- Recent industry work at Delta focused on tactile/visual policy learning, robot-learning infrastructure, contact-rich manipulation, and emerging world-model directions for real-world decision-making.
- Academic work at NTU and Columbia connects trustworthy multimodal reasoning and evaluation with embodied AI, alongside current research at NUS on harmful video content detection and adversarial robustness.

Experience

Research Fellow Sep 2026–Present
National University of Singapore Singapore

- Research harmful video content detection and the robustness of multimodal language models to adversarial attacks, including prompt injection, and investigate defenses.

Senior Associate Researcher 2025–Aug 2026
Delta Robotics Innovation Center, Delta Electronics Taiwan

- Built end-to-end robot-learning pipelines for manipulation, from multimodal data collection and curation to VLA-style policy training, rollout evaluation, and on-robot validation.
- Investigated tactile and visual policy learning for contact-rich manipulation, including failure analysis for slip, misalignment, under-force, and observation mismatch.
- Developed embodied-AI prototypes that connect language, perception, planning, action, and emerging world-model-oriented reasoning for robust real-world execution.

Postdoctoral Researcher 2023–2025
National Taiwan University (NTU) Taiwan

- Led and mentored projects on embodied navigation, mobile manipulation, policy monitoring, false-premise navigation, and affordance-aware planning.
- Built evaluation and dataset pipelines for embodied AI, including benchmark design, failure-mode analysis, and zero-shot VLM-based replanning workflows.
- Led projects resulting in CoRL 2024, EMNLP 2024, and ACL 2026, and mentored student-led robotics work accepted at NeurIPS 2024, AAAI 2026, and ACL 2026.

Visiting Researcher 2022–2023
Columbia University, hosted by Prof. Shih-Fu Chang New York, NY

- Developed early zero-shot causal video reasoning frameworks and maintained a strong bridge between embodied AI and trustworthy multimodal reasoning.
- Initiated and coordinated a Columbia–NTU collaboration that led to an EMNLP 2024 publication.

Selected Robot Learning Publications

Full publication history: Google Scholar.

- **VLN-NF: Feasibility-Aware Vision-and-Language Navigation with False-Premise Instructions** *(ACL 2026)* [paper] [project page] — Evidence-seeking embodied navigation with explicit abstention under false-premise or infeasible instructions. **Role: Lead**
- **Context-Aware Replanning with Pre-Explored Semantic Map for Object Navigation** *(CoRL 2024)* [paper] [code] — Zero-shot visual map refinement and semantic replanning for more robust embodied navigation. **Role: Lead**
- **AED: Adaptable Error Detection for Few-shot Imitation Policy** *(NeurIPS 2024)* [paper] — Policy monitoring for few-shot imitation learning with deployment-oriented error detection. **Role: Mentor**
- **Affordance-Guided Coarse-to-Fine Exploration for Base Placement in Open-Vocabulary Mobile Manipulation** *(AAAI 2026)* [paper] — Training-free VLM-guided planning for mobile-manipulation base placement in open-vocabulary settings. **Role: Mentor**
- **ADAPT: Benchmarking Commonsense Planning under Unspecified Affordance Constraints** *(ACL 2026)* [paper] — Benchmarking and evaluating planning under underspecified affordance and feasibility constraints. **Role: Mentor**
- **Unveiling Narrative Reasoning Limits of Large Language Models with Trope in Movie Synopses** *(EMNLP 2024)* [paper] [code] — Analyzes narrative reasoning limits and chain-of-thought hallucinations in LLMs using movie tropes. **Role: Lead**
- **Stage Conscious Attention Network (SCAN) : A Demonstration-Conditioned Policy for Few-Shot Imitation** *(AAAI 2022)* [paper] — Few-shot imitation learning through stage-aware policy conditioning from limited demonstrations. **Role: Mentor**
- **A Coarse-To-Fine (C2F) Representation for End-To-End 6-DoF Grasp Detection** *(CoRL 2020)* [paper] — 6-DoF grasp detection in cluttered scenes with a coarse-to-fine representation for manipulation.

Education

PhD in Computer Science 2023
National Taiwan University (NTU), Advisor: Prof. Winston H. Hsu Taiwan

- Dissertation: *Multi-modal Video Comprehension and Beyond*
- Visiting researcher at Columbia University, hosted by Prof. Shih-Fu Chang.

MS in Computer Science 2017
National Taiwan University Taiwan

BS in Computer Science 2013
National Tsing Hua University Taiwan

Skills

- **Robot Learning / Embodied AI:** imitation learning, diffusion policies, VLA-style policies, world-model-oriented planning, policy monitoring, failure analysis, and deployment-oriented evaluation.
- **Manipulation / Navigation / Planning:** mobile manipulation, base placement, embodied navigation, affordance-aware planning, evidence-seeking exploration, and false-premise instruction handling.
- **Multimodal Sensing:** RGB-D, 3D point clouds, tactile/contact feedback, open-vocabulary perception, and sensor-driven failure analysis.
- **Systems / Robotics Stack:** Python, PyTorch, ROS2, Linux, Docker, Git, Isaac Sim, and real-robot rollout/data pipelines.

Academic Services

- Area Chair: NAACL/COLING 2027 (ARR 10/2026); ACL 2026 (ARR 01/2026); EACL 2026 (ARR 10/2025).
- Reviewer: ICML, NeurIPS, CVPR, ICCV, NAACL, ACL, EMNLP, CoRL, ICRA, IROS, AAAI, ACM Multimedia, TCSVT.

References

References are available upon request.